Every step shown | **File** 7 / 7

ANS

Chapter 2 full worked answers

الإجابات الكاملة: افتحه بعد حل ملف الأسئلة

Each answer gives the result, the reason a marker looks for, and in the `why` lines the trap the question was built on. Numeric answers were computed by code from the question data.

Part A · Concept and interpretation (multiple choice)

Give the letter and a one-line reason.

A1 2-1 CLO2-B

(c)

Width = UCL − LCL + 1 = 44 − 35 + 1 = 10 (or the next LCL − this LCL = 45 − 35). The value 9 is the classic trap.

A2 2-1 CLO2-B

(c)

42/6 = 7 exactly, but the width is rounded **up** even from a whole number. With width 7, six classes cover only 18–59 and miss the value 60.

A3 2-1 CLO2-B

(d)

A class with no values stays in the table with f = 0.

A4 2-1 CLO2-B

(b)

f = RF × n = 0.15 × 60 = 9.

A5 2-2 CLO3-A

(c)

Midpoints for polygons; boundaries for histograms; upper boundaries for ogives.

A6 2-2 CLO3-A

(b)

The line starts at zero on the first lower class boundary.

A7 2-2 CLO3-B

(b)

The cumulative frequency does not increase across the class, so no values fall in it (the 52.5–60.5 drivers class).

A8 2-2 CLO3-B

(b)

Skew is named after the long tail, here on the right.

A9 2-3 CLO2-C

(c)

Nationality is qualitative (nominal), so a bar graph or pie chart. Histograms and ogives need quantitative classes.

A10 2-3 CLO2-C

(c)

0.15 × 360° = 54°.

A11 2-3 CLO3-B

(c)

A section's frequency is its top minus its bottom: 35 − 20.

A12 2-3 CLO3-A

(c)

Same place values as the key: stem tenths, leaf hundredths.

Part B · Short exam problems by lesson

Construction, completing tables, reading graphs, choosing graphs, find-the-mistake.

B1 2-1 CLO2-B

R = 47 − 12 = 35; width = 35/5 = 7, a whole number, but still rounded **up** to **8**. With width 7, the five classes would end at 46 and miss 47.

LIMITS	BOUNDARIES	F	RF	CF
12–19	11.5–19.5	5	0.167	5
20–27	19.5–27.5	9	0.3	14
28–35	27.5–35.5	9	0.3	23
36–43	35.5–43.5	4	0.133	27
44–51	43.5–51.5	3	0.1	30

Check: Σf = 30 = n ✓. The last class reaches 51 ≥ 47 ✓.

B2 2-1 CLO2-B

f_2 = 11 − 3 = **8**; CF_3 = 11 + 12 = **23**; f_4 = 36 − 23 = **13**; f_5 = 40 − 36 = **4**. RF: 0.075, 0.2, 0.3, 0.325, 0.1 (sum 1).

Below 39.5 = CF of the third class: 23/40 = **57.5%**.

B3 2-1 CLO2-B

Limits: **16–20, 21–25, 26–30, 31–35** (add and subtract 0.5). Width **5** (= 20.5 − 15.5). Midpoints **18, 23, 28, 33**.

B4 2-1 2-3 CLO2-B · CLO3-B

MAJOR	F	%	DEGREES
Statistics	9	22.5	81°
Mathematics	12	30	108°
Physics	6	15	54°
Chemistry	8	20	72°
Biology	5	12.5	45°
Total	40	100	360°

Mathematics is the most common major (30%), while Biology is the least (12.5%).

B5 2-2 CLO3-B

(a) 50 − 28 = **22 states**. (b) (48 − 10)/50 = **76%**. (c) **109.5–114.5** (18 states): the ogive's steepest segment.

B6 2-2 CLO2-C

(a) Midpoints with the extra empty classes:
(14.5, 0), (24.5, 4), (34.5, 9), (44.5, 15), (54.5, 7), (64.5, 5), (74.5, 0).

(b) Upper boundaries with CF, starting at the first lower boundary:
(19.5, 0), (29.5, 4), (39.5, 13), (49.5, 28), (59.5, 35), (69.5, 40).

B7 2-2 CLO3-B

(a) **bell-shaped** (b) **reverse J-shaped**: highest at the left and falling steadily (c) **U-shaped** (d) **bimodal** (e) **uniform**

B8 2-2 CLO2-C · CLO3-A

(1) The x-axis must show **boundaries** (99.5, 104.5, ...), not limits. (2) Histogram bars must **touch**. (3) Title and axis labels are required. (4) A frequency axis should start at 0. If an axis doesn't start at 0, mark the break with a zig-zag.

B9 2-3 CLO2-C

MODE	F	%	DEGREES
Bus	14	35	126°
Car	18	45	162°
Walk	5	12.5	45°
Taxi	3	7.5	27°

Car and bus: 32/40 = 80%, i.e. 288°.

B10 2-3 CLO3-B

YEAR	MALES	FEMALES	TOTAL
1	14	25 − 14 = 11	25
2	18	34 − 18 = 16	34
3	22	40 − 22 = 18	40

Enrolment grows every year (25 → 40), and males outnumber females each year.

B11 2-3 CLO2-C · CLO3-B

```
1 | 3 5 8
2 | 0 2 2 4 7 9
3 | 1 3 4 4 6 8
4 | 1 5        key: 1 | 3 = 1.3 s
```

Leaf counts 3, 6, 6, 2: **roughly symmetric (mound-shaped)**, with the peak of **6** values at stems 2 and 3.

B12 2-3 CLO3-B

Upward trend: sales rose from 42 to 71 (+29), with a single dip in April. The fastest change was **May to June (+8)**, the steepest segment.

B13 2-2 2-3 CLO2-C

(a) **time series**: change over time (b) **bar graph**: categorical (c) **histogram** (or frequency polygon): quantitative shape (d) **pie**: parts of a whole (e) **ogive**: counts below a value

Part C · Long structured questions

Full table-to-graph workups. Marks in brackets.

C1 2-1 2-2 CLO2-B · CLO2-C · CLO3-B **[16 marks]**

1. R = 44 − 27 = 17; width = 2.833 → **3**.

LIMITS	BOUNDARIES	MIDPOINT	F	RF	CF	CRF
27–29	26.5–29.5	28	2	0.05	2	0.05
30–32	29.5–32.5	31	3	0.075	5	0.125
33–35	32.5–35.5	34	11	0.275	16	0.4
36–38	35.5–38.5	37	14	0.35	30	0.75
39–41	38.5–41.5	40	7	0.175	37	0.925
42–44	41.5–44.5	43	3	0.075	40	1

2. Modal class **36–38** (f = 14). The frequencies 2, 3, 11, 14, 7, 3 rise to one central peak and fall away: **roughly bell-shaped**.

3. CF at 35.5 = **16 days**, i.e. 40%.

4. Histogram: boundaries 26.5, 29.5, ..., 44.5. Polygon: midpoints 25 (extra), 28, 31, 34, 37, 40, 43, 46 (extra). Ogive: 26.5 (CF 0), then each upper boundary.

C2 2-1 2-2 CLO2-B · CLO3-B **[10 marks]**

1. Width **10**; limits **30–39, 40–49, 50–59, 60–69, 70–79**; n = **40**.

2. Midpoints 34.5, 44.5, 54.5, 64.5, 74.5; RF 0.10, 0.175, 0.325, 0.25, 0.15.

3. (4 + 7 + 13)/40 = **60%**.

4. **49.5–59.5**: it holds the largest frequency (13), so the CF climbs fastest there.

C3 2-1 2-3 CLO2-B · CLO2-C · CLO3-C **[12 marks]**

PLACE	F	%	DEGREES	MALES	FEMALES
Library	21	35	126°	9	12
Home	18	30	108°	11	7
Café	9	15	54°	4	5
Lab	12	20	72°	6	6
Total	60	100	360°	30	30

3. Grouped: two bars side by side (male, female) for each place. Stacked: one bar per place of total height f, split into a male section and a female section on top.

4. **The library is the most popular place (35%)**; **home is preferred more by males (11 vs 7), the library more by females (12 vs 9)**.

Part D · True / false with justification

The justification is the mark.

D1 2-1 CLO2-B · CLO3-A

False. Classes must be continuous, so the empty class stays with f = 0.

D2 2-1 CLO2-B · CLO3-A

False. Round up even from a whole number, to 7. Otherwise the largest value falls outside the last class.

D3 2-2 CLO2-B · CLO3-A

False. Cumulative frequency never decreases. It is flat when a class is empty.

D4 2-2 CLO2-B · CLO3-A

False. A histogram shows quantitative classes on a number line with touching bars; a bar graph shows categories with separated bars.

D5 2-3 CLO2-B · CLO3-A

True. Every value can be read back from its stem and leaf, which a histogram cannot do.

D6 2-3 CLO2-B · CLO3-A

True. The parts must make up the whole circle.

D7 2-2 CLO2-B · CLO3-A

False. Every bar is divided by the same n, so only the y-scale changes.

D8 2-1 CLO2-B · CLO3-A

False. That gives 9 for the class 20.5–28.5, whose width is 8. Use UCB − LCB, or UCL − LCL + 1.

Part F · In-Class Quiz I simulation, Chapter 2 part (15 marks · 15 minutes)

Closed book, calculator allowed.

F1 2-1 CLO2-B

45/5 = 9, a whole number, so rounded up to **10**. Classes **47–56 (46.5–56.5)** and **57–66 (56.5–66.5)**.

F2 2-1 CLO2-B

CF_3 = **32**; CRF_3 = **0.8**; RF_4 = 8/40 = **0.2**.

F3 2-2 CLO3-A

Histogram: boundaries **119.5–129.5**; polygon: midpoint **124.5**; ogive: upper boundary **129.5**.

F4 2-3 CLO2-C

27/120 = **22.5%**; 0.225 × 360 = **81°**.

F5 2-3 CLO3-B

n = 2 + 4 + 7 + 4 + 1 = **18**; peak **7 values in the 60s**; **roughly bell-shaped (symmetric)**.

Chapter 2 study set: 7 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's slide pack, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief · 01 · Lesson 2-1: Organizing Data · 02 · Lesson 2-2: Histograms, Frequency Polygons, Ogives · 03 · Lesson 2-3: Other Types of Graphs · 04 · Revision pack · 05 · Exam-style questions · **06 · Full worked answers**